Interesting operators (say T) on Euclidean space V (later: Hermitian space too)

① Orthogonal operator: $\boxed{\langle Tv, Tw\rangle = \langle v, w\rangle \quad \forall\, v, w \in V}$

Let A = matrix of T wrt a chosen O.N. basis $e_1, \dots, e_n$

Rmks · Such T preserves norm (so $Tv = 0 \Rightarrow v = 0$, so T is an isomorphism)

· Ex Linear T with $\|Tv\| = \|v\|\ \forall\, v \Rightarrow T$ is orthogonal

· Let matrix of T wrt an O.N basis be A. Then T orthogonal $\overset{\text{Ex}}{\iff} A^TA = I$ (Check!)

$\iff$ cols of A are orthonormal wrt dot product

· Note $A^TA = I \iff A^{-1} = A^T \iff AA^T = I \iff$ rows of A are orthonormal too!

· $A^TA = I \xRightarrow{\det} \det(A^T)\det(A) = 1 \Rightarrow \det(A)^2 = 1 \Rightarrow \det(A) = \pm 1$

Ex v an evector of orthogonal T. Then $(Tv = \lambda v \Rightarrow |\lambda| = 1)$

- O_n = set of all orthogonal operators / matrices

$A, B \in O_n \Rightarrow AB \in O_n$ & $A^{-1} = A^T$

O_n = orthogonal group

SO_n = special ortho group

- $O_n \supset SO_n = \{A \in O_n \mid \det A = 1\}$ is a subgroup: closed under mult & inverses
- Ex Show $O_n = SO_n \amalg M(SO_n)$ where M is any fixed orthogonal matrix with det -1 e.g. $M = \begin{bmatrix} -1 & 0 \\ 0 & I \end{bmatrix}$
- Thm: $O_2 = \left\{ \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \mid a^2+b^2=1 \right\} \amalg \left\{ \begin{bmatrix} a & b \\ b & -a \end{bmatrix} \mid a^2+b^2=1 \right\}$

Thm (Euler): SO_3 = set of "rotations" in $\mathbb{R}^3$ (Ex.)
↓ meaning?

SO_n rotation by θ where $\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$

↳ reflection in the angle bisector of rays $\mathbb{R}^+ e_1$ and $\mathbb{R}^+ \begin{bmatrix} a \\ b \end{bmatrix}$ } check!

② Symmetric operator T on Eucl. space is defined by property $\boxed{\langle Tv, w\rangle = \langle v, Tw\rangle \ \forall\, v, w \in V}$

9 am tomorrow

Meaning for matrix A wrt O.N. basis

From $\langle Te_i, e_j\rangle = \langle e_i, Te_j\rangle$ we get $A_{ij} = A_{ji}$

i.e. T symmetric $\iff$ { Matrix of T wrt ON. basis is symmetric

note $\langle v, w\rangle = [v]^T[w] = [v]\cdot[w]$

coordinate vectors

$\langle Tv, w\rangle = (A[v])^T[w] = [v]^T A^T[w]$

$\langle v, Tw\rangle = [v]^T A\,[w]$ // For symmetric T

rmk $e_i^T M e_j = M_{ij}$

A Spectral thm. For any symmetric operator T on a Euclidean space V, there is an orthonormal basis of V consisting of eivectors of T "T can be orthogonally diagonalized"

Matrix version: A symmetric real matrix $\Rightarrow$ there is an orthogonal matrix P such that $P^T A P = D$.

Pf strategy: ① Find an evector w. $\rightsquigarrow$ ② Show $(\mathbb{R}w)^\perp$ is T-invariant (easy check!)

↑ Requires an ideal!

Our solution will be to develop inner product spaces over $\mathbb{C}$!

Hermitian space!

(then $T|_{(\mathbb{R}w)^\perp}$ is automatically symmetric)

Use induction!

We will have spectral thms for following operators

symmetric, Hermitian, Unitary, normal

$\mathbb{R}$ $\qquad$ $\mathbb{C}$